Slide 1: The Problem — The Smallholder Poverty Trap

Why 30 Million Farmers Cannot Sell Outside Local Village Middlemen

- **86% of Indian landholdings are small or marginal** (< 2 ha, Agriculture Census 2015-16).
- **Estimated ~30 Million farmers rely solely on 2G button phones** without internet access or apps.
- **The Minimum Order Quantity (MOQ) Trap:** Harvests are 150-400 kg; commercial trucks require 2-5 tonnes.
- **Information Asymmetry:** Distress village sales. Mandya ragi modal was ■3,565/qtl vs MSP ■5,205/qtl.
- **Digital Divide:** Existing agri-tech apps require smartphones, 4G data, and English/formal reading literacy.

Speaker Notes: *Speaker Notes (Anirudh): 86% of farmers hold under 2 ha. A harvest of 3 quintals cannot hire a truck to Mandya mandi. They sell to village middlemen at dictated cash prices. Annadata Dhwani brings the market directly to their phone keypad.*

Slide 2: How It Works — Voice & Keypad Marketplace

From Missed Call to Pooled Truckload in 6 Steps

- **1. Zero-Cost Access:** Farmer gives a missed call (toll-free, zero data).
- **2. Narrowband Audio (8 kHz):** Instant automated callback in native Kannada or Hindi.
- **3. Simple Keypad DTMF:** Farmer keys in Crop (1=Ragi), Quantity (qtl), and Minimum Price (■).
- **4. Hyperlocal Pooling:** Engine aggregates micro-lots into 2-5 Tonne commercial clusters at FPO yard.
- **5. FPO Agent Verification:** Field agent verifies quality, moisture, and captures geotagged photo.
- **6. Institutional Bidding:** Verified cluster auctioned to buyers; farmer confirms winning bid via phone.

Speaker Notes: *Speaker Notes (Anirudh): The farmer dials our number and hangs up. The platform rings back. A recorded voice in Kannada guides them. Micro-lots are pooled at local collection points, verified by agents, and sold in bulk.*

Slide 3: Live Demonstration Cue

Real-Time End-to-End Transaction & Failure Resiliency

- **Scene 1 (Farmer Call):** Live missed call from button phone → callback received → Kannada/Hindi IVR.
- **Scene 2 (Fair Price Guardrail):** Farmer keys ■3,200/qtl → Warned against mandi floor ■3,430 → Farmer raises price.
- **Scene 3 (Reliability / Drop Recovery):** Call disconnected mid-entry → Re-dial continues seamlessly where stopped.
- **Scene 4 (FPO Verification):** Cluster hits 2 tonnes → Agent uploads live geotagged verification on PWA.
- **Scene 5 (Auction & Confirmation):** Institutional buyers bid live on WebSocket dashboard → Winning bid confirmed.

Speaker Notes: *Speaker Notes (Anirudh): We demonstrate this live today. Watch the Ops screen. Hitesh will give a missed call, trigger a key-bounce, simulate a network drop, and we will verify the pooled lot live on stage.*

Slide 4: Engineering Proof — Designed for Rural Telephony Physics

Deterministic Signalling Over Narrowband Telephony

- **DTMF over Speech Recognition:** 0% Word Error Rate (WER) on numbers. Eliminates tractor noise/dialect failures.
- **Narrowband Audio Pipeline:** Bandpass filtered (300-3400 Hz), 8 kHz mono 16-bit PCM, loudnorm to -16 LUFS.
- **Debounce & Key-Bounce Filtration:** Mechanical switch bounce dropped at 120 ms; idempotency prevents dupes.
- **Transactional State Integrity:** PostgreSQL row-level locks prevent race conditions; Redis cache is disposable.
- **Data Privacy by Design:** Farmer mobile numbers masked with E.164 tokens across all APIs.

Speaker Notes: *Speaker Notes (Anirudh): Rural voice AI fails when speech recognition encounters an 8 kHz phone line. We engineered around the physics: DTMF tone signalling for numbers has zero error rate.*

Slide 5: Unit Economics & Sustainable Viability

Positive Unit Margins From Day One

- **Cluster Transaction Value:** 24 quintals @ ■3,500 = ■84,000.
- **Buyer Facilitation Fee (1.5%):** ■1,260 gross revenue per closed cluster.
- **Telephony Direct Cost:** ~■26 per cluster (callback costs ~■1.56 per listing vs ■157.50 fee share).
- **FPO Field Verification Incentive:** ■50 per inspected cluster.
- **Net Contribution Margin:** ■1,184 per cluster (~94% contribution margin).
- **Fixed Cost Break-Even:** Exotel plan (■9,999/5mo) + DLT (■5,900) = Break-even at only 14 closed clusters.

Speaker Notes: *Speaker Notes (Anirudh): The unit economics are robust. On a 24-quintal cluster at 1.5% fee, revenue is ■1,260 against ■76 in direct telephony and agent costs. Break-even requires only 14 clusters in 5 months.*

Slide 6: Production Realities & Operational Architecture

What is Real, What is Simulated, and What is Paid in Production

- **Real in Production:** Core IVR state machine, real Agmarknet price feed, 324 synthesized 8kHz audio clips, Postgres row-lock bidding.
- **Real Telephony Pilot:** Tested on live Exotel trial account with physical handset callbacks.
- **Simulated for Hackathon:** Institutional payment settlement (direct FPO escrow), buyer concurrency test bots.
- **Paid Operating Requirements:** Commercial Exotel trunking, DLT SMS entity registration, APMC trade compliance.
- **Future Roadmap:** Offline PWA sync for remote field agents, automated weight bridge IoT integration.

Speaker Notes: *Speaker Notes (Anirudh): We are transparent about the engineering: the engine, audio pipeline, and market pricing are 100% production-grade. The payment settlement is simulated for this demo.*